



Teheran University of Medical Sciences

International campus Vice-Dean for Educational Affairs

No. 124, Baradaran-e-Mozafar St.
Dameshgh St., Vahid-Asr Ave.
Tehran 3439123900 Iran.
Tel: (+98) 21-88 89 36 78
Web: gsia.tums.ac.ir/en/icedu
Email: icedu@tums.ac.ir

Final Exam
International Campus, DDS / Pharma D./MD Program
Biochemistry
September 2022 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Nourozi

Date of Exam: Jan 29, 2023

Duration of Exam: 50 Minutes

MCQ Question	50
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



Which of the following is synthesized In Phosphogluconate pathway :		1
NADH	الف	
FADH	ب	
NADPH	ج	
GTP	د	

Which of the following is a high energy molecule?		2
dihydroxyacetone-P	الف	
glyceraldehydes 3- P	ب	
1,3-biphosphoglycerate	ج	
2 phosphoglycerate	د	

Which of the following enzymes is important in fermentation?		3
LDH	الف	
ALT	ب	
AST	ج	
ALP	د	

Which of the following enzymes is important in glycogenolysis?		4
Phosphorylase	الف	
Hexokinas	ب	
Glucokinase	ج	
Phosphatase	د	

Which glucose transporter is active in liver tissues?		5
GLUT-1	الف	
GLUT-2	ب	
GLUT-3	ج	
GLUT-4	د	

All the following enzymes are important in gluconeogenesis except:		6
PEPCK	الف	
Fructose 1,6 biphosphatase	ب	
Glucose 6 phosphatase	ج	
Phosphorylase	د	

How many pyruvate molecules are required for gluconeogenesis?		7
1	الف	
2	ب	
3	ج	
4	د	

Where does glycolysis take place?		8
Cytosol	الف	
Mitochondria	ب	
Nucleus	ج	
Cell membrane	د	

Oxidation of one glucose molecule yields how many ATP molecules?		9
26-28	الف	
30-32	ب	
34-36	ج	
38-40	د	

For cellulose synthesis which molecule is important in glucose activation?		10
ADP	الف	
CDP	ب	
GDP	ج	
TDP	د	



		11
Which molecule is synthesized in HMP shunt?	الف	
NAD	ب	
NADH	ج	
FADH	د	
NADPH		

		12
How many ATP molecules are synthesized in glycolysis?	الف	
1	ب	
2	ج	
3	د	
4		

		13
HMG-CoA reductase activity is increased by the administration of the hormone:	الف	
Insulin	ب	
Epinephrine	ج	
Glucagon	د	
Glucocorticoids		

		14
Anti-inflammatory corticosteroids inhibit...	الف	
Phospholipase A1	ب	
Lipo-oxygenase	ج	
Phospholipase A2	د	
Cyclo-oxygenase		

		15
Genetic deficiency of lipoprotein lipase causes hyperlipoproteinemia of the following type:	الف	
Type I	ب	
Type IIa	ج	
Type IIb	د	
Type V		



Acetyl CoA formed from pyruvate can be used for the synthesis of all the following except:		16
Cholesterol	الف	
Fatty acids	ب	
Steroid hormones	ج	
Glucose	د ✓	

NADPH is produced when this enzyme acts.		17
Pyruvate dehydrogenase	الف ✓	
Malic enzyme	ب	
Succinate dehydrogenase	ج	
Malate dehydrogenase	د	

maybe blue ??

Long-chain fatty acyl CoA esters are transported across the mitochondrial membrane by.....		18
cAMP	الف	
Choline	ب ✓	
Prostaglandin	ج	
Carnitine	د ✓	

During each cycle of β -oxidation of fatty acid, all the following compounds are generated except:		19
NADH	الف	
H ₂ O	ب	
FADH	ج	
Acyl CoA	د ✓	

Acetyl CoA required for fatty acid synthesis is produced by:		20
Pyruvate dehydrogenase complex	الف	
Citrate lyase	ب	
Thiolase	ج ✓	
Carnitine-acyl transferase	د	



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No. 124, Sarafshar-e-Musavi
Carmenhigh St., Vahid Ave.
Tehran 3439123908 Iran.
Tel: (+98) 21-48 89 36 79
Web: gata.tums.ac.ir/meduicedu
Email: icedu@tums.ac.ir

All statements regarding 3-OH-3 methyl glutaryl CoA are true except:		21
It is formed in the cytoplasm ?	الف	
Required in ketogenesis	ب	
Involved in the synthesis of Fatty acid	ج	
An intermediate in cholesterol biosynthesis	د	

Which of the following is an activator of fatty acid synthesis?		22
Palmitate	الف	
Carnitine ✓	ب	
Malonyl CoA	ج	
Citrate	د	

stearic Acid $\Rightarrow$ 146

1 molecule of palmitic acid in total oxidation to Co ₂ will yield molecules of ATP (as high energy bonds):		23
119	الف	
129	ب	
146	ج	
154	د	

β -Oxidation of odd-carbon fatty acid chain produces:		24
Propionyl CoA	الف	
Malonyl CoA	ب	
Succinyl CoA	ج	
Glutaryl CoA	د	

In the respiratory chain, cytochrome a3 is located in complex.....		25
I	الف	
II	ب	
III	ج	
IV	د	

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Another way of this MCQ that came in Kish

or has the lowest tendency to oxidize?

Which part of the respiratory chain has the lowest oxidizing power?			26
FMN (Flavin Mononucleotide)	الف		
Coenzyme Q	ب		
Cytb	ج		
Cyte	د		

The ATP/ADP transporter in the mitochondrial inner membrane is inhibited by			27
Oligomycin	الف		
CN ⁻	ب		
Atractyloside	ج		
Barbiturates	د		

Which of the complexes of the respiratory chain does not act as a proton pump?			28
Succinate Q Reductase	الف		
NADH dehydrogenase	ب		
Cytochrome C oxidase	ج		
Cyt C reductase	د		

Deficiency or hereditary absence of which enzyme causes immunodeficiency in infants?			29
Thymidylate synthetase	الف		
Adenosine deaminase	ب		
Xanthine oxidase	ج		
Ribonucleotide reductase	د		

All the following compounds are involved in the biosynthesis of pyrimidine nucleotides except:			30
Aspartate → for Purine synthesis	الف		
Phosphoribosyl pyrophosphate	ب		
Carbamyl phosphate	ج		
Glycine	د		

if it said biosynthesis of Purine, then the answer would be Aspartate



In humans the end product of purine catabolism is...		31
Urea	الف	
Xanthine	ب	
Uric acid	ج	
Allantoin	د	

Lesch-Nyhan syndrome, the sex-linked recessive disorder is due to the lack of the enzyme:		32
Hypoxanthine-guanine phosphoribosyl transferase	الف	
Adenosine deaminase	ب	
Xanthine oxidase	ج	
Adenine phosphoribosyl transferase	د	

In which of following hyperbilirubinemias (jaundices), the amount of conjugated bilirubin in the blood would increase the most?		33
Obstructive	الف	
Hepatitis	ب	
Hemolytic	ج	
Physiologic	د	

The first nucleotide formed during purine de novo synthetic pathway is		34
PRPP	الف	
Uric acid	ب	
OMP	ج	
IMP	د	

Which enzyme is inhibited by Allopurinol to treat which disease?		35
Orotate decarboxylase, Renal Azotemia	الف	
Xanthine oxidase; Gout	ب	
Orotate decarboxylase, Gout	ج	
Xanthine oxidase; SCID	د	

↳ is also came with just the Enzyme

..... donates its phosphate to ADP to yield ATP during muscle exertion.		36
Arginine glycine amidinotransferase	الف	
Creatine phosphate	ب	
Creatine kinase	ج	
Guanidoacetate	د	

All the following are nonfunctional enzymes <u>except</u> :		37
Alkaline phosphatase	الف	
Acid phosphatase	ب	
Lipoprotein lipase	ج	
Gamma glutamyltranspeptidase	د	

Which among the following is <u>not</u> a pancreatic specific enzyme?		38
Lipase	الف	
Aldolase	ب	
Trypsin	ج	
Amylase	د	

Which of the following is <u>the only</u> plasma enzyme normally found in urine?		39
Acid Phosphatase	الف	
Nucleotidase	ب	
Nucleotidase	ج	
Amylase	د	

Which among the following is <u>not</u> a pancreatic specific enzyme?		40
Lipase	الف	
Aldolase	ب	
Trypsin	ج	
Amylase	د	



Which of the following statements about the urea cycle is correct?		41
Argininosuccinate is lysed to urea and ornithine in the urea cycle.	الف	
Carbamoyl phosphate supplies both of the nitrogen atoms of urea in the urea cycle.	ب	
Arginine is hydrolysed to urea and ornithine in the urea cycle. ?	ج	
The formation of urea from the urea cycle yields energy.	د	

Which of the following statements about transamination reactions is correct?		42
Transamination reactions involve ATP hydrolysis.	الف	
Transamination reactions are irreversible.	ب	
Transamination reactions require NAD^+ or NADP^+ .	ج	
Transamination reactions require pyridoxal-5'-phosphate. ?	د	

Which of the following amino acids cannot provide a substrate for gluconeogenesis?		43
Leucine	الف	
Tryptophan	ب	
Histidine	ج	
Isoleucine	د	

Which of the following is non-essential amino acid?		44
Lysine	الف	
Serine	ب	
Leucine	ج	
Threonine	د	

The amino acid which serves as a carrier of ammonia from skeletal muscle to liver is:		45
Alanine	الف	
Methionine	ب	
Arginine	ج	
Glutamine	د	

COUNTINUE TO THE NEXT PAGE....



Amino acids are released into the portal system by:		46
Pinocytosis	الف	
Active transport	ب	
Facilitated transport ?	ج	
Passive transport	د	

Which of the following enzyme catalyze the major mechanism for transporting ammonia in a nontoxic form?		47
Aminotransferase	الف	
Amino acid oxidase	ب	
Glutamine synthetase	ج	
Threonine dehydrogenase	د	

Which of the following enzyme is deficient in alkaptonuria?		48
Homogentisate dioxygenase	الف	
p-Hydroxyphenyl- pyruvate dioxygenase	ب	
Phenylalanine hydroxylase	ج	
Tyrosinase	د	

A person with phenylketonuria cannot convert ...		49
Phenylalanine to tyrosine	الف	
Phenylalanine to isoleucine	ب	
Phenol into ketones	ج	
Phenylalanine to lysine	د	

An example of a transamination process is		50
glutamate = hexanoic acid + NH ₃	الف	
aspartate + hexanoic acid = glutamate + oxaloacetate	ب	
aspartate + α ketoglutarate = glutamate + oxaloacetate	ج	
glutamate = α-ketoglutarate + NH ₃	د	

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